Sampling, Curricula, and Partial Rollouts

Deep Dive into Kimi K3 — Chapter 20

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Foundation Books Project

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Goal: score difficulty per snapshot, separate target from acquisition, preserve unfinished work through typed pauses, admit only complete groups, and hand the learner per-segment behavior scores — each Kimi claim in its evidence class.

Roadmap

Choosing tasks

Preserving unfinished work

Admitting complete groups

Choosing tasks

Difficulty, target, and acquisition

- Typed, admitted tasks only; difficulty is versioned: counts $(9, 1), (5, 5), (1, 9)$ give raw $(1/10, 1/2, 9/10)$, Beta$(1, 1)$-smoothed $(1/6, 1/2, 5/6)$.

[DERIVED: chapter studio fixtures] [REPORTED: k1.5 Sec. 2.3.4: $1 - s_i$ priority]

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- Target $p=(1/3,1/3,1/3)$ is the objective; acquisition $q=(1/15,1/3,3/5)$ is the observation law. Weights $w=(5,1,5/9)$ give every $q_i w_i = 1/3$; for $f=(1,0,-1)$ the raw mean $-8/15$ corrects to 0.

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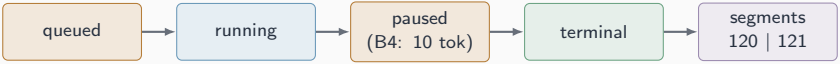
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- Support $p(i)>0 \Rightarrow q(i)>0$ is necessary; clipping at two biases the mean to $-1/5$.

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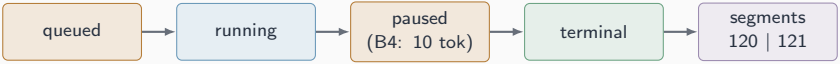
Preserving unfinished work

Pause, lifecycle, and four meanings of reuse



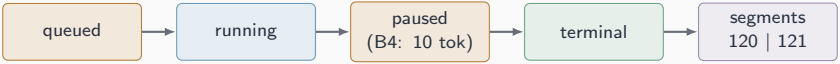
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- Reuse $R = (c, d, \ell, o)$: prefix $(1, 0, 0, 0)$, suffix $(0, 0, 1, 1)$; 48 tokens generated, 38 in the base loss.

Admitting complete groups

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Diagnose the earliest broken interface. Next: Chapter 21's Multi-Domain and Reasoning-Effort Reinforcement Learning.